

minimum variance portfolios sound like great ideas: Who doesn't want to maximize risk-adjusted returns or minimize volatility? However, look at the difference in recommended portfolio weights over the two time periods! For the tangency portfolio, from 1927 to 1960, the model recommends an 88 percent allocation to long bonds; however, from 1961 to 2013 the model recommends a 44 percent allocation. Correlation matrices matter!

[Table 6.2](#) Tangency and Minimum Variance Portfolio Weights

1927–1960	Tangency Weights	Min. Var. Weights
W(SP500)	17.92%	–0.55%
W(HML)	–5.81%	7.07%
W(LTR)	87.90%	93.48%
1961–2013	Tangency Weights	Min. Var. Weights
W(SP500)	41.22%	18.37%
W(HML)	15.27%	38.95%
W(LTR)	43.51%	42.68%

Remarkably, we find that a simple, equal-weighted allocation outperforms both of these “optimized” portfolios in tests of varying time periods (similar to the results from DeMiguel, Garlappi, and Uppal, 2009). [Table 6.3](#) tabulates the results. The strategies represented are as follows: a strategy that allocates equally to the three asset classes each month (equal weights); a strategy that uses the tangency portfolio weights recommended from 1927 to 1960 (tangency); and a strategy that uses the recommended minimum variance weights estimated from 1927 to 1960 (minimum variance). Since we know that time periods matter, we analyze our results from 1961 to 2013 to see *if the strategy is robust*. We present performance metrics for compound annual growth rates, worst drawdowns, Sharpe ratios, and Sortino ratios (see [Table 6.3](#) for numbers and see the Appendix for the definitions of the various statistics). Based on risk-adjusted metrics, the simple equal-weight strategy outperforms the complex asset allocation algorithm (highest Sharpe/Sortino ratios).